

Case Study #4:

Research Project Management Office (RMPO)

Main Focus of the Case Study

This case study focuses on the management of research project data at Kookle University. The RMPO (Research Project Management Office) is responsible for maintaining and tracking information about research projects, grants, donors, and the faculty, staff, and students involved.

OBJECTIVE

1. Maintain Data with CRUD method

Administrators can retrieve, review, and update information within the Research Project Management Office (RMPO) system by using standard CRUD (Create, Read, Update, Delete) operations to ensure accurate and up-to-date database management.

2. Analyse and report statistics for management

Administrators can utilize the data stored in the system to perform statistical analysis and generate reports. These insights can serve as reliable references to support decision-making and strategic management.

3. Tracking, planning and forecasting

Administrators can track the progress of each research project and leverage project data for planning purposes. The system also allows forecasting of project timelines, required resources, or budgets to improve efficiency and project outcomes.

4. Research performance of each faculty

Administrators can evaluate the research performance of each faculty or department, enabling the assessment of capabilities, achievements, and overall contributions to institutional research success.

Kookle University
Project Initiation Form

Date: 1 Jan 2019
Project Title: Ontology-based Plant Disease Detection System
Project Description: The project aims to ...

Keywords: Ontology, Plant Disease, Intelligent System

Project Category: ☐ Research and Development
☐ Training/Capacity Building
☐ Conference/Workshop

Research Areas of the Project: Semantic Web, Ontology, Knowledge Representation

Project Duration: 12 months

Contract Period: 1 Jan 2019 – 31 Dec 2019

Budget: 1,000,000 THB

Installment:

No.	Date	%Budget	Amount
1	30 April 2019	30	300,000
2	31 August 2019	30	300,000
3	31 December 2019	40	400,000

Principle Investigator:

1. Name: Mark Zuckerberg Email: mark@kmail.com
Department ICT Department Area of Expertise: Social Analytics, AI
2. Name: Elon Musk Email: musk@kmail.com
Department ICT Department Area of Expertise: Knowledge Representation

Project Member:

1. Name: Tim Berners Lee Email: tim@kmail.com
Department ICT Department Area of Expertise: Semantic Web Ontology
2. Name: _____ Email: _____
Department _____ Area of Expertise: _____

Donor/Sponsor Name: Thailand Research Fund

Donor Type: ☐ Research Organization
☐ International Organization
☐ Private Company

Donor Contact Info: Mr. Somchai N. CMMU Building Phayathai Bangkok 10400 Thailand
Tel: (+66) 2 343 1500

Student Assistant:

1. Name: Nina Billa Email: nina@kmail.com
Department: ICT
Responsibility: Programming
Maximum working hours per month: 40 Rate per hour: 300
Start Date: 1 Feb 2019 End date: 30 June 2019
2. Name: Lucy Dell Email: lucy@kmail.com
Department: Business Management
Responsibility: Project Coordinator
Maximum working hours per month: 40 Rate per hour: 300
Start Date: 1 Jan 2019 End date: 31 Dec 2019

Based on the form, the requirements for the Kookle University Research Project Management Office (RPMO) system are as follows:

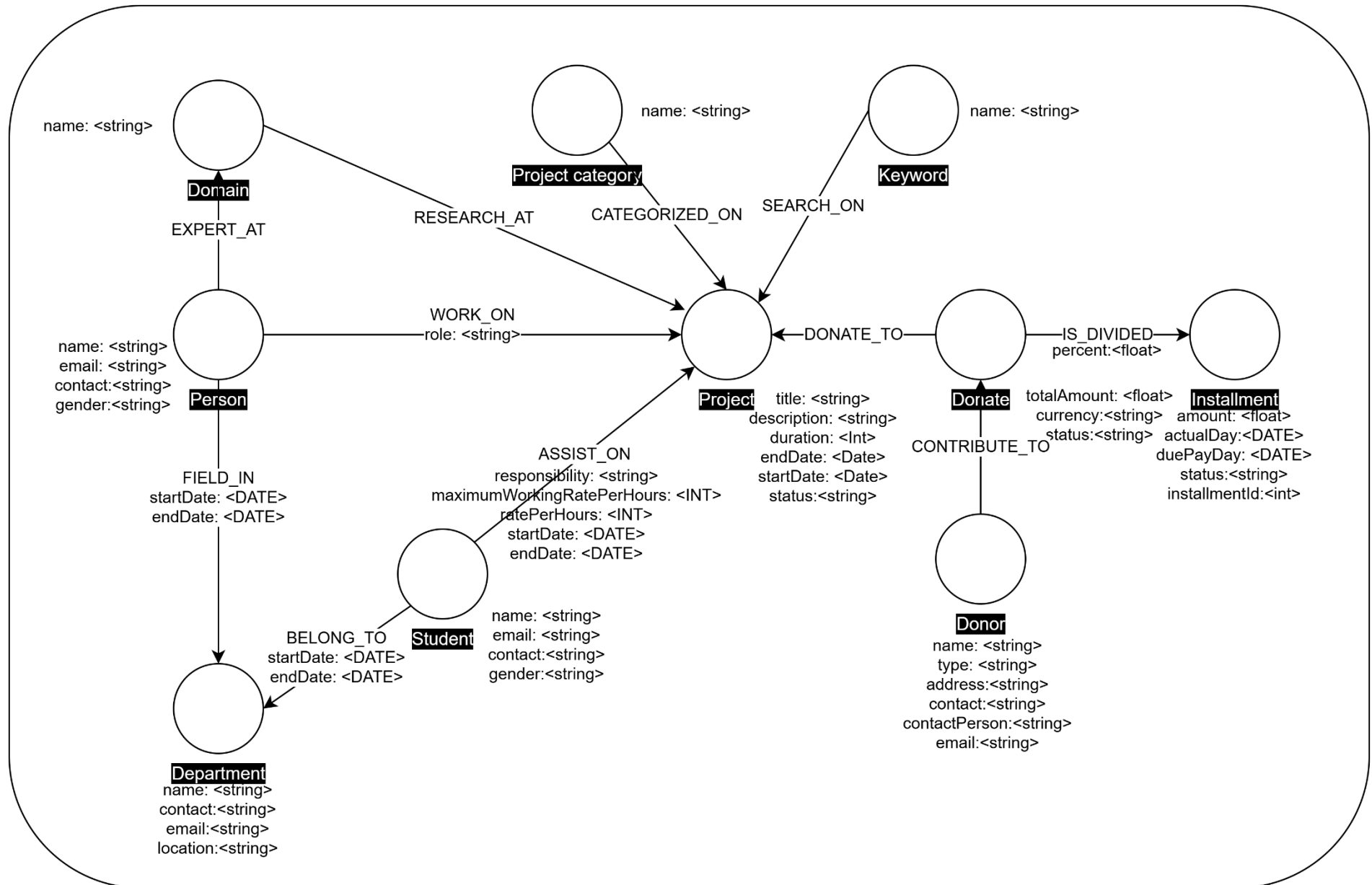
Data Maintenance: The system must maintain data about research projects, grants received, donors, and the faculty, staff, and students working on the projects.

Data Entry and Updates: It should support the entry and updating of project-related data after a project form is submitted.

Querying: The system needs to allow for querying data from project spreadsheets. This includes queries to coordinate with the project investigator.

Reporting: The system should be able to generate useful reports or statistics for university management. Examples of reports include total grants received for each active research area, statistics about sponsors/donors, and the research performance of each faculty/staff member.

Graph- conceptual diagram



QUERY

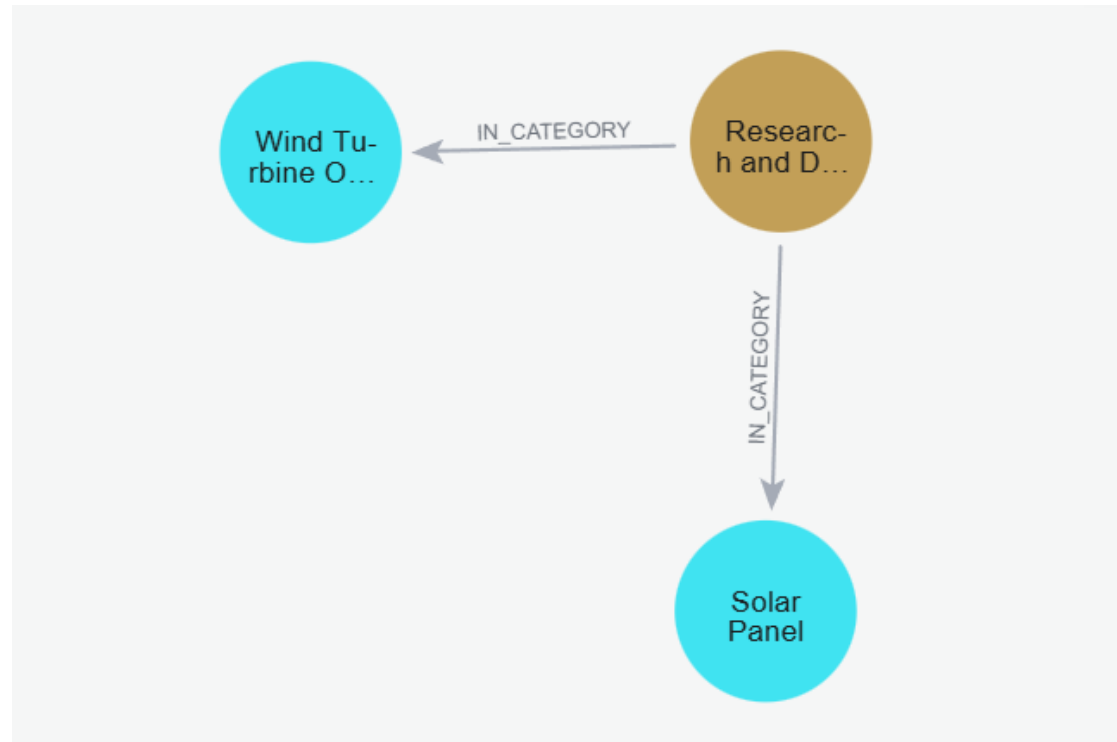
1. Find all active projects in "Research and Development" category

Cypher Command:

```
MATCH (cat:Category)-[l:IN_CATEGORY]->(proj:Project)
```

```
WHERE proj.status = 'Active' AND cat.name = 'Research and Development'
```

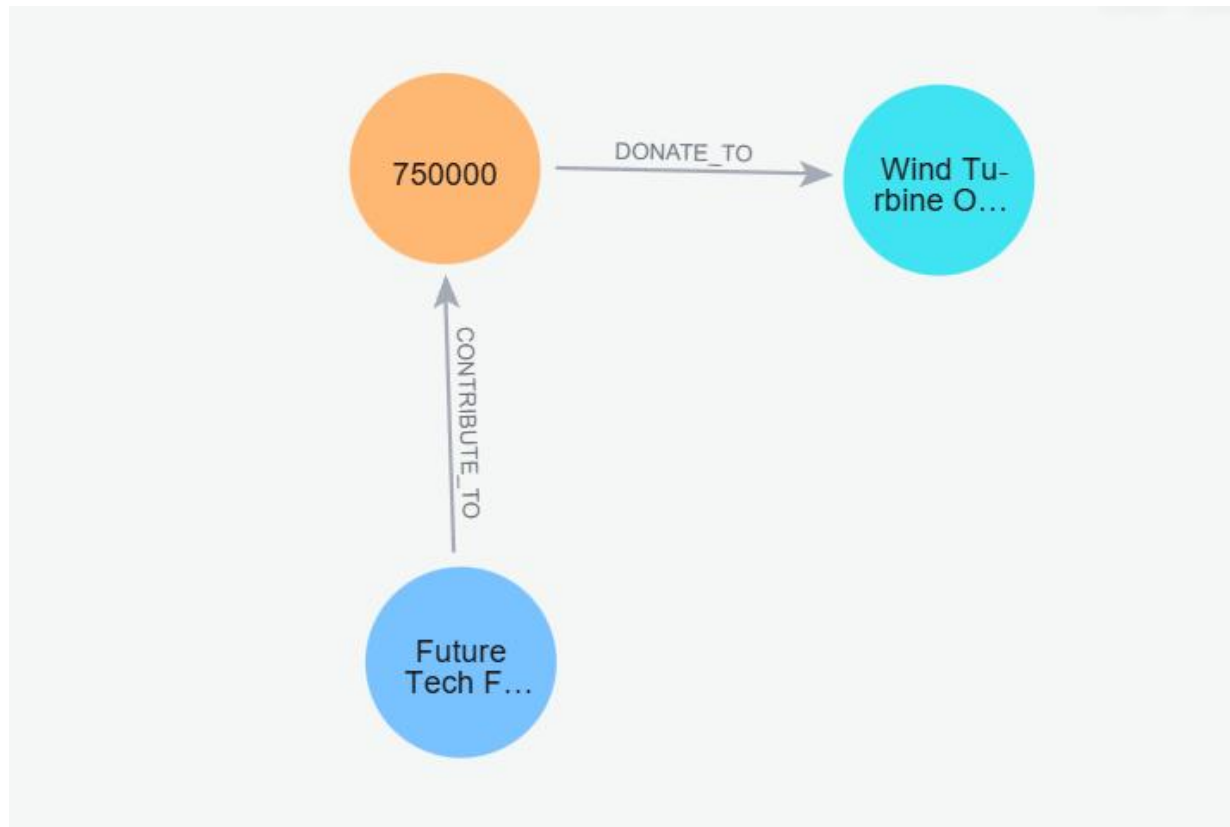
```
RETURN proj, cat, l
```



2. List all projects funded by donors of type " Private Company "

Cypher Command:

```
MATCH p=(donor:Donor {type: 'Private Company'})-[:CONTRIBUTE_TO]->(don:Donate)-[:DONATE_TO]->(proj:Project)
RETURN p
```



3.Calculate total budget amount by research area for current year

Cypher Command:

WITH

date({year: 2025, month: 1, day: 1}) AS startOfYear,

date({year: 2025, month: 12, day: 31}) AS endOfYear

MATCH (d:Domain)-[:RESEARCH_AT]->(p:Project)<-[:DONATE_TO]-(don:Donate)

WHERE p.startDate <= endOfYear AND p.endDate >= startOfYear

RETURN

d.name AS ResearchArea,

SUM(don.totalAmount) AS TotalBudget,

don.currency AS Currency

ORDER BY TotalBudget DESC

	ResearchArea	TotalBudget	Currency
1	"Renewable Energy"	1700000	"THB"
2	"Data Science"	1250000	"THB"
3	"Geology"	1200000	"THB"
4	"Aerodynamics"	750000	"THB"
5	"Material Science"	750000	"THB"

4. Show project details with donor information and installment schedule

Cypher Command:

```
MATCH (donor:Donor)-[:CONTRIBUTE_TO]->(d:Donate)-[:DONATE_TO]->(p:Project)
```

```
MATCH (d)-[:IS_DIVIDED]->(i:Installment)
```

```
WITH p, d, donor, i ORDER BY i.duePayDay
```

```
RETURN
```

```
p.title AS Project,
```

```
donor.name AS Donor,
```

```
d.totalAmount AS TotalDonation,
```

```
collect({
```

```
id: i.installmentId,
```

```
due: i.duePayDay,
```

```
amount: i.amount,
```

```
status: i.status
```

```
}) AS InstallmentSchedule
```

	Project	Donor	TotalDonation	InstallmentSchedule
1	"Solar Panel"	"Global Philanthropy Inc."	500000	[{ amount:100000, due:2023-03-01, id:"I-001", status:"Paid" }, { amount:200000, due:2023-09-01, id:"I-004", status:"Paid" }, { amount:200000, due:2024-03-01, id:"I-005", status:"Pending" }]
2	"Wind Turbine Optimization"	"Future Tech Foundation"	750000	[{ amount:250000, due:2024-03-15

5. Identify projects ending within next 3 months

Cypher Command:

```
MATCH (proj:Project)
WHERE proj.endDate >= date()
AND proj.endDate < date() + duration('P3M')
RETURN
proj.title AS Project,
proj.endDate AS EndDate,
proj.status AS Status
```

	Project	EndDate	Status
1	"Wind Turbine Optimization"	2026-01-01	"Active"

6. Calculate average project duration by category

Cypher Command:

```
MATCH (c:Category)-[:IS_CATEGORIZED]->(p:Project)
```

```
RETURN
```

```
  c.name AS Category,
```

```
  avg(p.duration) AS AverageDurationInMonths
```

Category

AverageDurationInMonths

```
1  "Research and D  28.0
   evelopment"
```



7. List all international organization donors and their contact details

Cypher Command:

MATCH (d:Donor)

WHERE d.type = 'International Organization'

RETURN d as DonorDetail

DonorDetail

```
(:Donor {address: "123 Charity Lane, Washington DC", contact: "+1-202-555-0199", name: "Global Philanthropy Inc.", contactPerson: "Jane Smith", type: "Inte
```

8. Calculate total funding received from each donor type by year

Cypher Command:

```
MATCH (d:Donor)-[:CONTRIBUTE_TO]->(don:Donate)-[:IS_DIVIDED]->(i:Installment)
```

```
WHERE i.actualDay IS NOT NULL
```

```
RETURN
```

```
  d.type AS DonorType,
```

```
  i.actualDay.year AS Year,
```

```
  don.currency AS Currency,
```

```
  SUM(i.amount) AS TotalReceived
```

```
ORDER BY
```

```
  DonorType, Year
```

	DonorType	Year	Currency	TotalReceived
1	"International Organization"	2023	"THB"	300000
2	"Private Company"	2024	"THB"	250000

9. Identify top 5 research areas by total funding received

Cypher Command:

```
MATCH (d:Domain)-[:RESEARCH_AT]->(p:Project)<-[:DONATE_TO]-(don:Donate)
```

```
RETURN
```

```
  d.name AS ResearchArea,
```

```
  SUM(don.totalAmount) AS TotalFunding,
```

```
  don.currency AS Currency
```

```
ORDER BY
```

```
  TotalFunding DESC
```

```
LIMIT 5
```

	ResearchArea	TotalFunding	Currency
1	"Renewable Energy"	1700000	"THB"
2	"Data Science"	1250000	"THB"
3	"Geology"	1200000	"THB"
4	"Aerodynamics"	750000	"THB"
5	"Material Science"	750000	"THB"

10. Show project timeline and payment schedule

Cypher Command:

```
MATCH (p:Project)<-[:DONATE_TO]-(d:Donate)-[:IS_DIVIDED]->(i:Installment)
```

```
RETURN
```

```
  p.title AS Project,
```

```
  p.startDate AS ProjectStart,
```

```
  p.endDate AS ProjectEnd,
```

```
  i.installmentId AS InstallmentID,
```

```
  i.duePayDay AS PaymentDueDate,
```

```
  i.amount AS PaymentAmount,
```

```
  i.status AS PaymentStatus
```

```
ORDER BY
```

```
  Project, PaymentDueDate
```

	Project	ProjectStart	ProjectEnd	InstallmentID	PaymentDueDate	PaymentAmount	PaymentStatus
1	"Geothermal Energy Plant"	2024-06-01	2027-06-01	"I-003"	2024-06-15	400000	"Pending"
2	"Geothermal Energy Plant"	2024-06-01	2027-06-01	"I-008"	2025-01-15	400000	"Pending"
3	"Geothermal Energy Plant"	2024-06-01	2027-06-01	"I-009"	2025-06-15	400000	"Pending"
4	"Solar Panel"	2023-01-15	2025-01-14	"I-001"	2023-03-01	100000	"Paid"
5	"Solar Panel"	2023-01-15	2025-01-14	"I-004"	2023-09-01	200000	"Paid"
6	"Solar Panel"	2023-01-15	2025-01-14	"I-005"	2024-03-01	200000	"Pending"
7	"Wind Turbine Optimization"	2024-01-01	2026-01-01	"I-002"	2024-02-15	250000	"Paid"
8	"Wind Turbine Optimization"	2024-01-01	2026-01-01	"I-006"	2024-08-15	250000	"Pending"

11. List projects with their total budget and percentage of budget received

Cypher Command:

```
MATCH (p:Project)-[:DONATE_TO]-(d:Donate)-[:IS_DIVIDED]->(i:Installment)
```

```
WHERE i.actualDay IS NOT NULL
```

```
WITH p, d, COALESCE(SUM(i.amount), 0) AS amountReceived
```

```
RETURN
```

```
p.title AS Project,
```

```
d.totalAmount AS TotalDonation,
```

```
amountReceived AS AmountReceived,
```

```
round((toFloat(amountReceived) / toFloat(d.totalAmount)) * 100, 2) AS CompletionPercentage
```

```
ORDER BY
```

```
CompletionPercentage DESC
```

	Project	TotalDonation	AmountReceived	CompletionPercen
1	"Solar Panel"	500000	300000	60.0
2	"Wind Turbine Optimization"	750000	250000	33.33

12. Update project status when contract end date is reached

Cypher Command:

```
MATCH (p:Project)
```

```
WHERE p.endDate <= date() AND p.status <> 'Completed'
```

```
SET
```

```
p.status = 'Completed',
```

```
p.actualEndDate = p.endDate
```

```
RETURN
```

```
p.title AS UpdatedProject,
```

```
p.status AS NewStatus,
```

```
p.actualEndDate AS CompletionDate
```

	UpdatedProject	NewStatus	CompletionDate
1	"Solar Panel"	"Completed"	2025-01-14

13. Insert new installment record for approved project payments

Cypher Command:

```
CREATE (i:Installment {
  installmentId: 'I-010',
  duePayDay: date('2026-01-10'),
  amount: 50000,
  status: 'Pending',
  actualDay: null
})

WITH i
MATCH (d:Donate)-[:DONATE_TO]->(p:Project {title: 'Solar Panel'})
SET d.totalAmount = d.totalAmount + 50000
CREATE (d)-[r:IS_DIVIDED {
  amount: 50000,
  date: date(),
  percent: 9.1
}]->(i)

RETURN p.title AS Project, d.totalAmount AS NewTotalBudget, i AS NewInstallment
```

	Project	NewTotalBudget	NewInstallment
1	"Solar Panel"	550000	(:Installment {amount: 50000, installmentId: "I-010", duePayDay: 2026-01-10, status: "Pending"})

14. Delete project title = "Solar Panel" and associated records

Cypher Command:

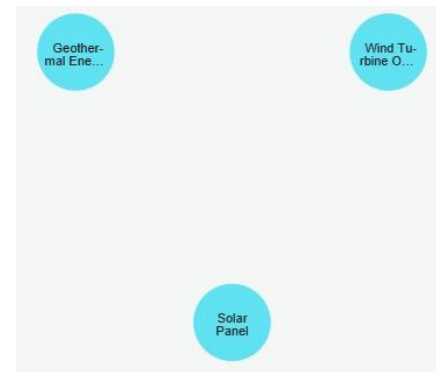
```
MATCH (p:Project {title: 'Solar Panel'})
```

```
OPTIONAL MATCH (p)-[:DONATE_TO]-(d:Donate)
```

```
OPTIONAL MATCH (d)-[:IS_DIVIDED]->(i:Installment)
```

```
DETACH DELETE p, d, i
```

Before



Implemented

✓ Deleted 6 nodes, deleted 18 relationships

After



15. Calculate monthly burn rate for active projects

Cypher Command:

```
MATCH (p:Project)
WHERE p.startDate <= date() AND p.endDate >= date()

OPTIONAL MATCH (p)<-[:DONATE_TO]-(d:Donate)

RETURN
  p.title AS Project,

  COALESCE(d.totalAmount, 0) AS TotalBudget,
  d.currency AS Currency,

  p.duration AS DurationInMonths,

  CASE
    WHEN p.duration IS NULL OR p.duration = 0 THEN 0.0
    ELSE round(toFloat(COALESCE(d.totalAmount, 0)) / p.duration, 2)
  END AS MonthlyBurnRate
```

	Project	TotalBudget	Currency	DurationInMonths	MonthlyBurnRate
1	"Wind Turbine Optimization"	750000	"THB"	24	31250.0
2	"Geothermal Energy Plant"	1200000	"THB"	36	33333.33

16. Track project keyword trends over the past 5 years

Cypher Command:

```
MATCH (k:Keyword)-[:SEARCH_ON]->(p:Project)
```

```
WHERE p.startDate >= date() - duration('P5Y') AND p.startDate <= date()
```

```
RETURN
```

```
  k.name AS Keyword,
```

```
  p.startDate.year AS Year,
```

```
  count(p) AS ProjectCount
```

```
ORDER BY
```

```
  Keyword, Year
```

	Keyword	Year	ProjectCount
1	"Budgeting"	2023	1
2	"Budgeting"	2024	1
3	"Drilling"	2024	1
4	"Energy Storage"	2023	1
5	"Energy Storage"	2024	1
6	"Machine Learning"	2024	2

17. For all ongoing projects, show the total number of installments, the number of incomplete installments, and the number of complete installments.

Cypher Command:

```
MATCH (p:Project)
```

```
WHERE p.endDate > date()
```

```
OPTIONAL MATCH (p)-[:DONATE_TO]-(:Donate)-[:IS_DIVIDED]->(i:Installment)
```

```
RETURN
```

```
  p.title AS Project,
```

```
  p.status AS Status,
```

```
  count(i) AS TotalInstallments,
```

```
  SUM(CASE WHEN i.actualDay IS NOT NULL THEN 1 ELSE 0 END) AS CompletedInstallments,
```

```
  SUM(CASE WHEN i.actualDay IS NULL THEN 1 ELSE 0 END) AS IncompleteInstallments
```

```
ORDER BY
```

```
  Project
```

	Project	Status	TotalInstallments	CompletedInstallments	IncompleteInstallments
1	"Geothermal Energy Plant"	"Planning"	3	0	3
2	"Wind Turbine Optimization"	"Active"	3	1	2

18. Identify projects that have passed their contract end date but are still 'On going'.

Cypher Command:

```
MATCH (p:Project)
```

```
WHERE p.endDate <= date() AND p.status <> 'Completed'
```

```
RETURN
```

```
  p.title AS OverdueProject,
```

```
  p.endDate AS ContractEndDate,
```

```
  p.status AS CurrentStatus
```

	OverdueProject	ContractEndDate	CurrentStatus
1	"Solar Panel"	2025-01-14	"Active"
			

19. For all ongoing projects, calculate the total estimated budget to be spent in each year (based on the contract end date)

Cypher Command:

```
MATCH (p:Project)<-[:DONATE_TO]-(d:Donate)
```

```
WHERE p.endDate > date()
```

```
WITH p, d, round(toFloat(d.totalAmount) / p.duration, 2) AS monthlyBurnRate
```

```
UNWIND range(0, p.duration - 1) AS monthOffset
```

```
WITH p, d, monthlyBurnRate, (p.startDate + duration({months: monthOffset})).year AS year
```

```
RETURN
```

```
  p.title AS Project,
```

```
  year AS Year,
```

```
  d.currency AS Currency,
```

```
  SUM(monthlyBurnRate) AS EstimatedAnnualSpend
```

```
ORDER BY
```

```
  Project, Year
```

	Project	Year	Currency	EstimatedAnnualS
1	"Geothermal Energy Plant"	2024	"THB"	233333.310000000006
2	"Geothermal Energy Plant"	2025	"THB"	399999.960000000014
3	"Geothermal Energy Plant"	2026	"THB"	399999.960000000014
4	"Geothermal Energy Plant"	2027	"THB"	166666.650000000002
5	"Wind Turbine Optimization"	2024	"THB"	375000.0

20. For each ongoing project, show the project lead (Principal Investigator), the total number of staff, and the project budget, to help in planning staff for new projects with similar budgets.

Cypher Command:

```
MATCH (p:Project)<-[:DONATE_TO]-(d:Donate)
```

```
WHERE p.endDate > date()
```

```
MATCH (lead:Person)-[:WORK_ON {role: 'Project Investigator'}]->(p)
```

```
MATCH (staff:Person)-[:WORK_ON]->(p)
```

```
WITH p, d, collect(DISTINCT lead.name) AS projectLeads, count(DISTINCT staff) AS totalStaff
```

```
RETURN
```

```
  p.title AS Project,
```

```
  projectLeads,
```

```
  totalStaff,
```

```
  d.totalAmount AS Budget,
```

```
  d.currency AS Currency
```

```
ORDER BY
```

```
  Budget DESC
```

	Project	projectLeads	totalStaff	Budget	Currency
1	"Geothermal Energy Plant"	["Alice", "Frank"]	4	1200000	"THB"
2	"Wind Turbine Optimization"	["Eve"]	3	750000	"THB"

21. Identify the areas of expertise with the fewest staff members in ongoing projects, for future recruitment planning

Cypher Command:

```
MATCH (p:Person)-[:WORK_ON]->(proj:Project)
```

```
WHERE proj.endDate > date()
```

```
WITH collect(DISTINCT p) AS activeStaff
```

```
MATCH (d:Domain)
```

```
OPTIONAL MATCH (d)<-[:EXPERT_AT]-(p:Person)
```

```
WHERE p IN activeStaff
```

```
RETURN
```

```
  d.name AS Domain,
```

```
  count(p) AS ActiveStaffCount
```

```
ORDER BY
```

```
  ActiveStaffCount ASC
```

	Domain	ActiveStaffCount
1	"Renewable Energy"	1
2	"Aerodynamics"	1
3	"Material Science"	1
4	"Data Science"	2
5	"Geology"	2

22. Identify available staff members (not currently part of an ongoing project) who can be assigned to new projects.

Cypher Command:

```
MATCH (p:Person)-[:WORK_ON]->(proj:Project)
WHERE proj.endDate > date()
WITH collect(DISTINCT p) AS unavailableStaff
MATCH (p:Person)
WHERE NOT p IN unavailableStaff
OPTIONAL MATCH (p)-[:EXPERT_AT]->(d:Domain)
RETURN
  p.name AS AvailableStaff,
  p.contact AS Contact,
  collect(d.name) AS Expertise
```

	AvailableStaff	Contact	Expertise
1	"David"	" +1-202-555-018	["Renewable Energy"]
		2 "	